Simranjeet Singh

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Platform Architect | Distributed Systems | Cloud & AI Platforms

Engineer with 17+ years building large-scale distributed systems and cloud platforms, operating at principal-level scope. Leads high-ambiguity, cross-functional initiatives spanning architecture, reliability, and developer platform foundations.

HIGHLIGHTS

- Owned intent resolution and the **EHR tool layer** (Epic, eClinicalWorks) of a **HIPAA-constrained voice agent** that books appointments and processes prescription refills.
- Shipped a read-only **DevOps triage agent** that investigates production alerts and reports probable cause.
- Designed the initial **RAG pipeline** for chat and voice.
- Architected **platform libraries** — circuit breaker, metrics, resilient clients, HIPAA-compliant auditing — adopted org-wide across multiple product teams.
- Led decomposition of a **15-domain modular monolith into 6 microservices** across 4 delivery phases.
- Separated **Kafka consumer and REST API workloads** across 9 services to eliminate GC contention and enable independent scaling.
- Designed **fair-queueing and background-processing** architectures for high-throughput multi-tenant workloads.
- Stabilized **JVM workloads at 40GB+ heaps** in production and established cluster-wide runtime standards.

AI-FIRST ENGINEERING

- Delivery built end to end on **agentic tooling and spec-driven development** — Claude Code, GitHub Copilot, OpenCode, CLI agents, BMAD, and OpenSpec.

EXPERIENCE

JPMorgan Chase - Lead Software Engineer (VP)

August 2026 - September 2026

- Architected and deployed a **5TB data migration pipeline** across AWS accounts using Glue, Spark, and Apache Airflow

Nextiva - Staff Engineer

March 2025 - July 2026

- Integrated **Stripe and Authorize.net** payment APIs with request-scoped idempotency keys, so retries and redeliveries dedupe without collapsing distinct customer charges.
- Separated **Kafka consumer workloads from REST API deployments** across 9 core services, eliminating shared GC pressure and thread-pool contention and enabling independent scaling of event-processing and API tiers.
- Architected a phased decomposition of a **15-domain modular monolith into 6 microservices** across 4 delivery phases, defining extraction sequencing, dual-write migration, and shared library boundaries.
- Architected **reusable platform libraries** (circuit breaker, metrics, resilient clients, HIPAA-compliant auditing, telemetry) and drove their adoption across multiple product teams.
- Designed the initial **RAG pipeline** across chat and voice — chunking, embedding strategy, hybrid search, and LLM synthesis.
- Built the **intent-to-tool layer** of a HIPAA-constrained voice agent — resolves caller intent, identifies the provider's EHR (eClinicalWorks or Epic), and calls its APIs to book appointments and process prescription refills; and built a read-only **DevOps triage agent** that investigates production alerts and reports probable cause.

OpenText - Software Architect

December 2021 - February 2025

- Designed **fair scheduling for multi-tenant execution** to prevent starvation and improve tenant fairness.
- Scaled data processing to **30TB/day** by architecting background-processing for large query workloads.
- Led the **SDM-to-FAS cloud migration**, aligning scope and delivery across the FAS and SDM engineering teams.
- Built the **.NET/C# on-prem collection agent's SDM processor** that enabled the SDM-to-FAS cloud migration.
- Introduced **distributed tracing primitives** (correlation IDs) to improve end-to-end diagnostics.

OpenText - Technical Lead

June 2019 - November 2021

- Improved **distributed job-scheduler behavior** for cancellation/delete-heavy workloads.
- Modernized **Kubernetes ingress stack** for better platform consistency.
- Introduced **Testcontainers-based integration testing** to improve confidence and maintainability.

OpenText - Technical Lead

January 2019 - June 2019

- Built a **custom Kafka connector for Vertica to RDBMS** synchronization to enable near real-time data flow and improved read performance.
- Developed **backend framework capabilities** for multi-dashlet support.
- Supported **release war-room performance triage** and rapid issue closure.

Earlier Roles

- Software Specialist, OpenText (2016-2019) · Member Technical Staff, Ness Technologies (2012-2016)
- Product Development Engineer, Symphony (2011-2012) · Project Engineer, Wipro Technologies (2008-2011)

TECHNICAL SKILLS

Languages: Java, C#, Groovy(Grails), TypeScript, Bash, Python, Go, SQL

AI/ML: GPT-4o, Gemini, LangChain, LangGraph, RAG (hybrid search + RRF)

AI/ML Infra: MongoDB Vector Search, Redis Vector Search, STT/TTS (Deepgram, Google Cloud TTS, NVIDIA NeMo)

AI-Assisted Development: Claude Code, GitHub Copilot, OpenCode, CLI agents

Spec-Driven Development: BMAD, OpenSpec

Frameworks: Spring, Hibernate, Angular, Flask, .NET

Cloud/DevOps: GCP, AWS, S3, Glue, Docker, Kubernetes, Argo CD, Jenkins, CI/CD, Prometheus, Grafana, ELK, Datadog

Data: BigQuery, PostgreSQL, MongoDB, Redis, Elasticsearch, Oracle, MySQL, MS-SQL, Vertica, Spark, Airflow

Messaging/Integration: Kafka, RabbitMQ, REST, gRPC

Security/Identity: OAuth2, SAML, JWT, SSL/TLS, OWASP practices

EDUCATION

B.Tech, Sastra University (2004 - 2008)